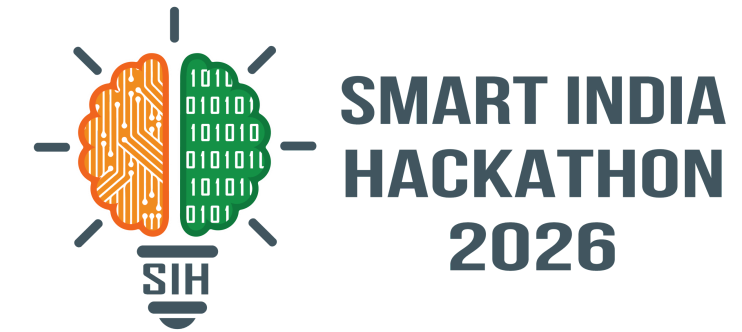
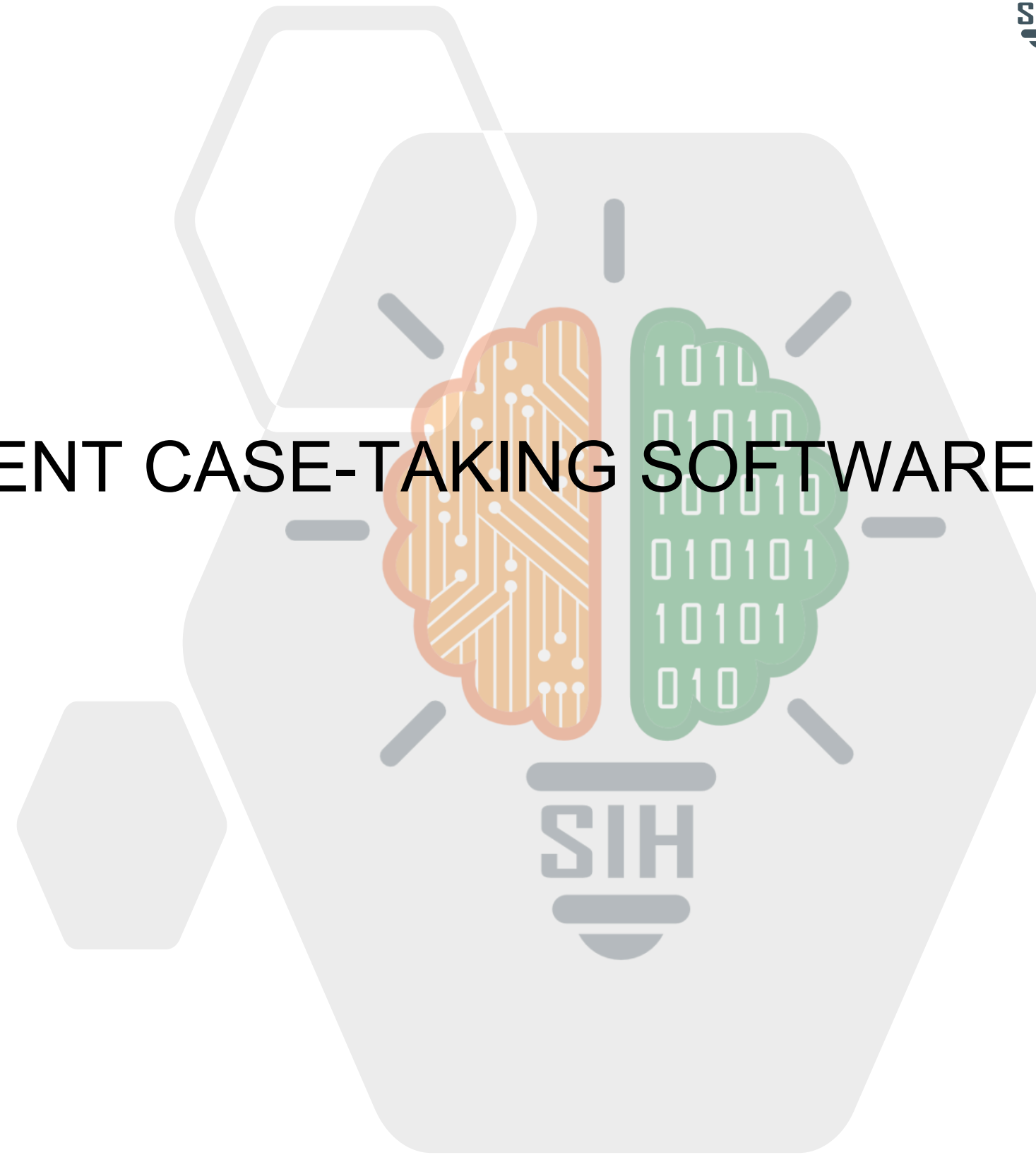


SMART INDIA HACKATHON 2026



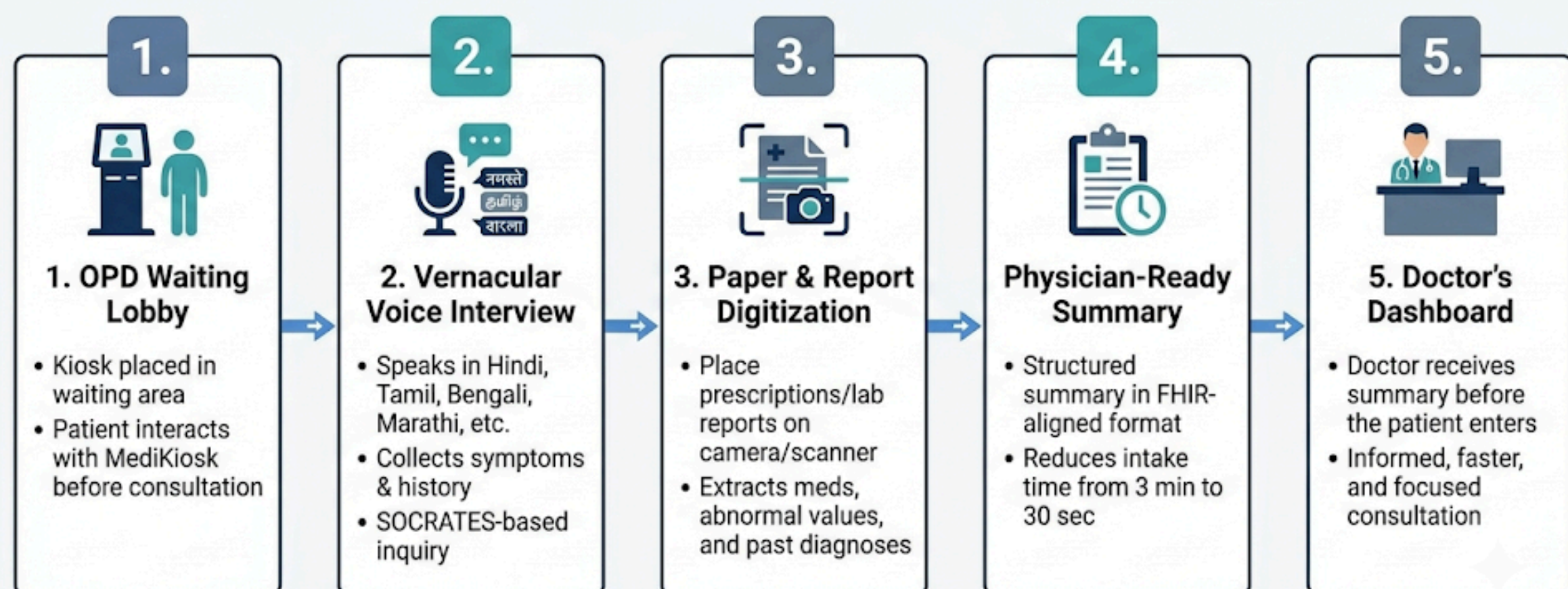
- **Problem Statement ID – 26047**
- **Problem Statement Title- PATIENT CASE-TAKING SOFTWARE**
- **Theme- HealthTech**
- **PS Category- Software**
- **Team ID - TEAM 020**
- **Team Name - HEXABYTE**



MediKiosk : An AI-Powered, Multilingual, Multimodal Clinical Pre-Consultation & Triage Kiosk for Indian Hospital OPDs.

Problem: India's government-hospital OPDs see 80–120+ patients per doctor per shift, leaving just 2–3 minutes per consultation. Much of this limited time is spent re-collecting patient history, handling paperwork, and overcoming language barriers, not diagnosing. As a result, critical symptoms go unheard, red flags are missed, and doctors are overwhelmed, putting timely and safe patient care at risk.

Proposed Solution



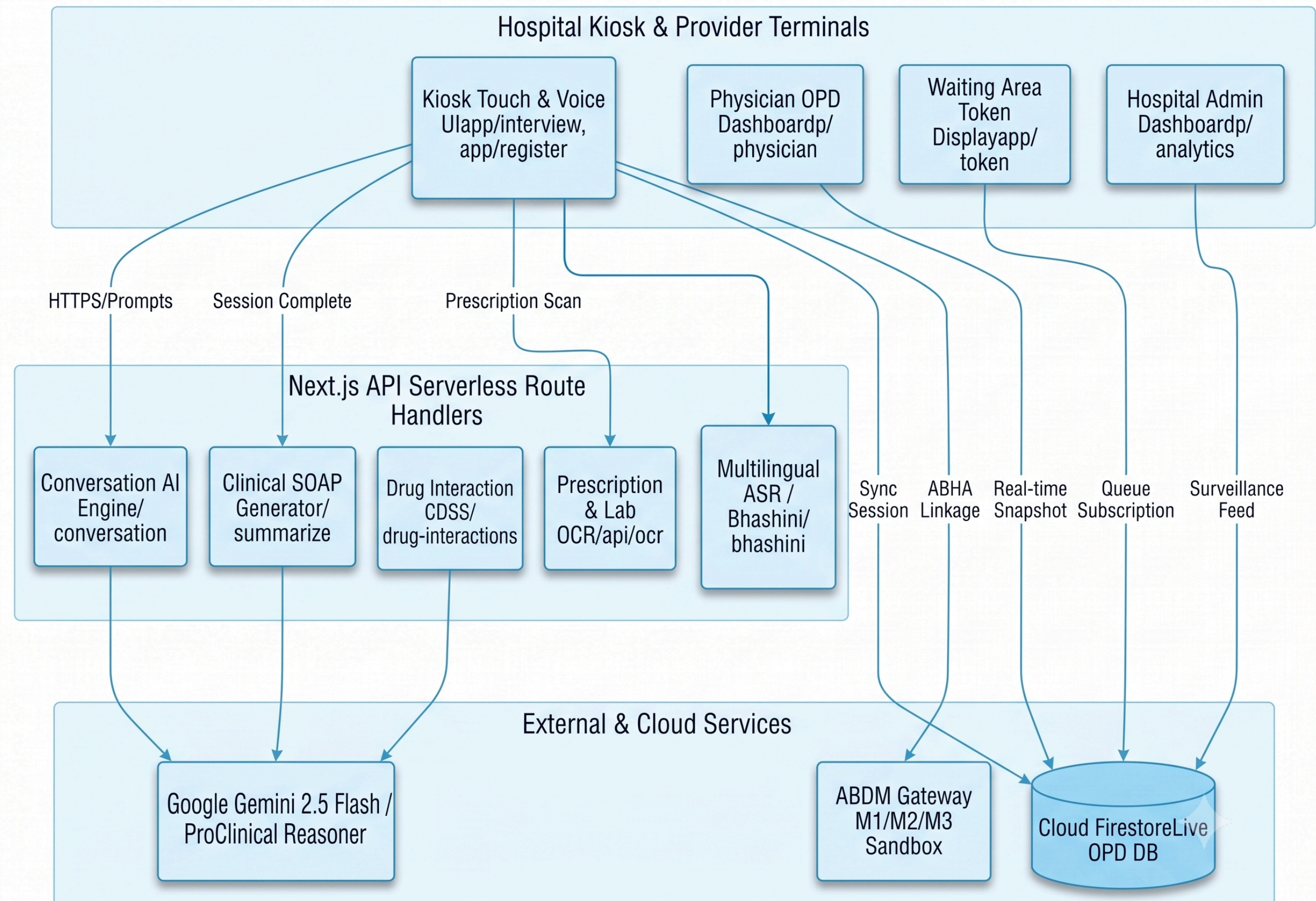
Innovation/Uniqueness

- **Hybrid voice-touch interface** with regional language binding sustains accuracy in high-noise OPD halls
- **Dual-tier red-flag engine** – instant keyword interception + contextual LLM risk-scoring – detects emergencies (chest pain, stroke, breathlessness) in real time and auto-escalates queue priority
- **Multi-pass vision extraction** cross-checks handwriting against Indian pharmacopeia, flagging low-confidence reads instead of guessing.
- **Follow-Up Reminders** – SMS & Email alerts 2 days prior.
- **OPD Token** – Unique token for queue placement.
- **OPD Queue** – Shows the live patient token queue
- **OPD Analytics** – Tracks patient flow and queue performance.



TECH STACK

- **Frontend** : Next.js 15 (App Router) + React 19
- **Audio & Accessibility** – Web Speech API with BCP-47 regional code binding (hi-IN, ta-IN, bn-IN...) + touch-chip fallback for high-noise OPD halls
- **Vision & Document Capture** – HTML5 Canvas image pre-processing with base64 compression, fed into Gemini Vision as a multimodal payload
- **AI Intelligence Engine** – AI intelligence engine: gemini-3.6-flash
- **Database** – Cloud-based Firebase with encrypted data transmission and secure decryption during retrieval.





Financial & Social Viability

- Massive addressable market, 25000+ government hospitals/PHCs plus large private OPD volume across
- Rides government policy tailwinds (Ayushman Bharat Digital Mission) a natural fit for public procurement, not a hard commercial sell.
- Cost is offset by throughput gains faster consultations mean more patients served per shift, a direct efficiency return.

Technical Feasibility

- Proven components: Indic ASR/TTS and prescription OCR are already production-grade, not experimental.
- Built-in fallback: voice+touch hybrid keeps working even if speech recognition fails in noisy OPDs.
- Standard hardware: just a touchscreen, camera, scanner and mic, nothing custom to build or manufacture.
- Multilingual by design: understands Hindi, Tamil, Bengali, Marathi, and more, not just English.



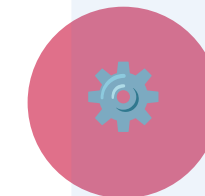
AI Hallucination

RISK

Wrong facts, wrong dosage

FIX

Doctor sign-off
required



OPD Bottlenecks

RISK

Noise, bad writing,
no net

FIX

Touch + offline
fallback



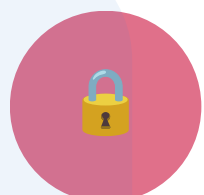
Missed Emergencies

RISK

Silent red flags

FIX

Instant two-tier
alerts



Data Privacy

RISK

Sensitive data misuse

FIX

Consent + zero-
retention



IMPACT

1. OPD Patients (Primary Users)

- Demographics: Rural and urban citizens attending public healthcare facilities, including non-literate and low health-literacy individuals.

2. Language Inclusion

- Supports multiple Indian languages, enabling patients from diverse backgrounds to communicate their symptoms accurately and confidently.

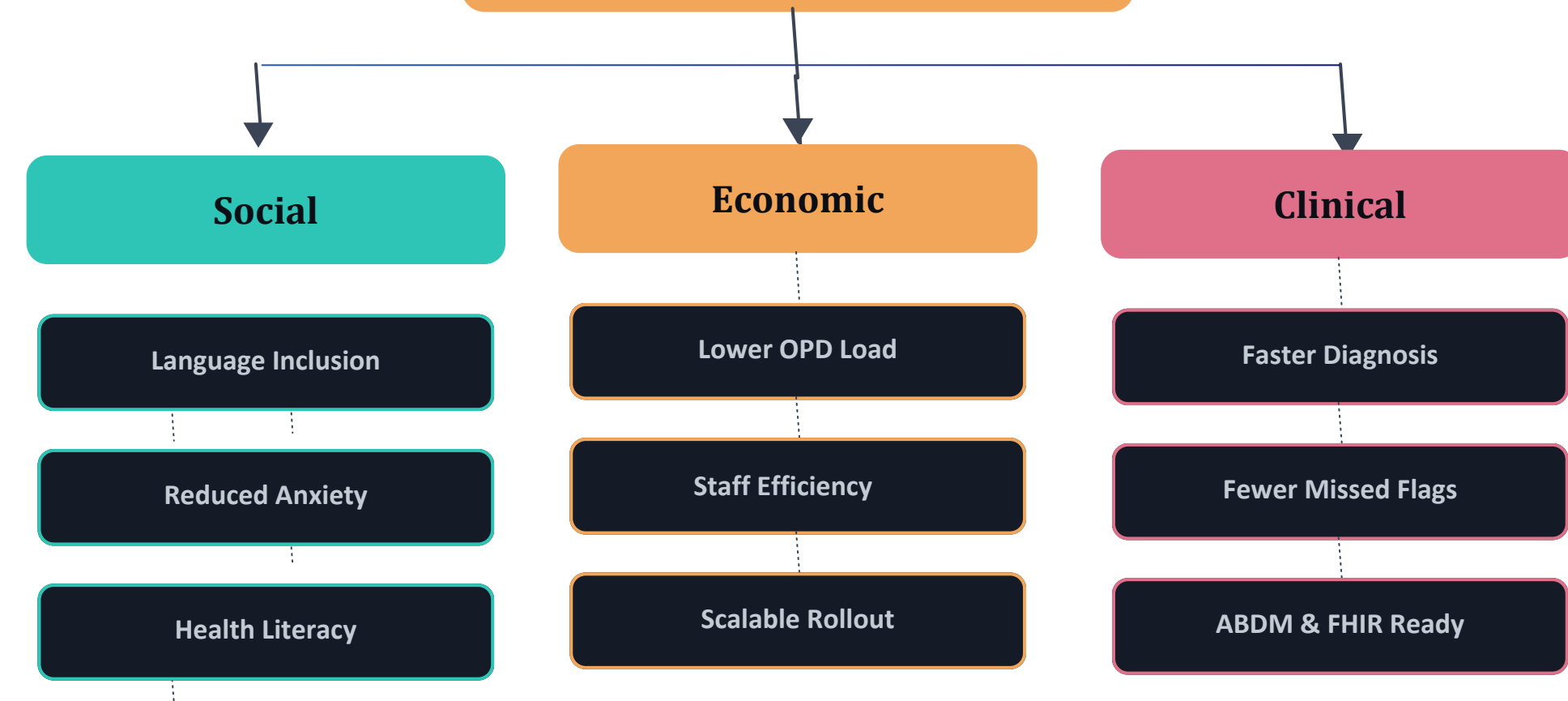
3. Improved Patient Engagement

- Interactive voice and touch-based communication encourages patients to actively participate in the consultation process and provide complete medical information.

4. Enhanced Healthcare Accessibility

- Makes quality healthcare services more accessible to rural, non-literate, and low health-literacy populations, reducing barriers to medical consultation.

Benefits of MediKiosk



- Improves healthcare accessibility by enabling multilingual, voice-touch interactions, helping patients communicate symptoms accurately regardless of literacy level.
- Enhances clinical efficiency through faster patient screening, early risk detection, and reduced workload for doctors and hospital staff.

Capability	MediKiosk	Agentic AI OPD Triage	SamaVaani (ASR Debias)	Protocol in Every Pocket	ASR Under the Stethoscope	AMIE (Google)
Multilingual voice interview	✓	~	~	✓	~	✗
Prescription digitization (vision)	✓	✗	✗	✗	✗	✗
Real-time red-flag detection	✓	~	✗	~	✗	✗
Doctor-ready structured summary	✓	✗	✗	✗	✗	✗

✓ Fully supported

~ Partial / limited

✗ Not present

Research

- SamaVaani: Auditing and Debiasing Multilingual Clinical ASR for Indian Languages
- Protocol in Every Pocket: AI at India's Healthcare Frontline (ORF)
- ASR Under the Stethoscope: Biases in Clinical Speech Recognition across Indian Languages
- Towards Conversational Diagnostic AI (AMIE)
-

References

<https://github.com/Inf3rno26/Patient-Case-Taking-Software-SIH202>
<https://medikiosk.vishadjain.in/>
Cloud Opus 3.6
Gemini 3.8